MTH622

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Q1.

Moment of inertia = 0.01 kg·m², angular velocity = 20 rad/s.

Kinetic energy will be ___ J.

- 1
- 5
- 4
- 2

Correct: 2

Q2.

Mass = 5 kg, radius = 0.1 m (solid sphere).

Moment of inertia about diameter will be ___ kg·m².

- 0.05
- 0.01

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- 0.02
- 0.10

Correct: 0.02

Q3.

If center of mass is at origin then coordinate axes and principal axes are:

- perpendicular
- coincident

Correct: coincident

Q4.

Let $\vec{\mathbf{r}} = 2\hat{\mathbf{i}}$ and $\vec{\mathbf{p}} = 3\hat{\mathbf{i}}$, magnitude of angular momentum will be:

- 1
- 0

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- 6
- 5

Correct: 0

Q5.

Moment of inertia = 0.44 kg·m², angular speed = 2 rad/s.

Rotational kinetic energy will be ___ J.

- 0.48
- 0.88
- 0.66
- 0.76

Correct: 0.88

Q6.

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For ellipsoid of inertia, if moment of inertia about OP is 0.25 kg·m², length of OP will be ___ m.

- 4
- 2
- 1
- 5

Correct: 1

Q7.

What happens to angular velocity when moment of inertia is doubled (angular momentum constant)?

- It is halved
- It doubles
- Remains same

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- It becomes zero

Correct: It is halved

Q8.

Mass = 2 kg, radius = 0.5 m (right circular cone).

Moment of inertia about axis of symmetry will be ___ kg·m².

- 0.50
- 0.35
- 0.15
- 0.42

Correct: 0.15

Q9.

Rate of change of momentum = 64 kg·m/s², acceleration = 16 m/s².

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Mass of particle will be ___ kg.

- 28
- 8
- 14
- 4

Correct: 4

Q10.

Mass = 1 kg, radius = 0.9 m (ring).

Moment of inertia about central axis will be ___ kg·m².

- 0.45
- 0.18
- 1.8
- 0.81

Correct: 0.81

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